David GILLIAM **MATH 1351 Spring 2004**

Homework Set HW9 due 3/20/04 at 11:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)
This is problem 12 Section 4.3 page 215.

Given $f(x) = \frac{1}{3}x^3 - 9x + 2$

(a) Find all critical numbers. If there are no such real x, type DNE in the answer blank. If there is more that one real x, give a comma separated list (e.g. 1,2). MAKE SURE TO ENTER THE NUMBERS IN INCREASING ORDER.

ANSWER (critical points) : ______

(b) Find where the function is increasing and decreasing.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______
ANSWER (Decreasing) : ______

(c) Find where the function is concave up and concave down.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (concave up): ______
ANSWER (concave down): ______

(d) Find the x coordinate all points of inflection. If there are no such real x, type DNE in the answer blank. If there is more that one real x, give a comma separated list (e.g. 1,2). MAKE SURE TO ENTER THE NUMBERS IN INCREASING ORDER.

ANSWER (inflection points): ______

2.(1 pt)
This is problem 15 Section 4.3 page 215.

Given $f(x) = x^3 + 35x^2 - 125x - 9375$

(a) Find all critical numbers. If there are no such real x, type DNE in the answer blank. If there is more that one real x, give a comma separated list (e.g. 1,2). MAKE SURE TO ENTER THE NUMBERS IN INCREASING ORDER.

ANSWER (critical points) : ______

(b) Find where the function is increasing and decreasing.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______
ANSWER (Decreasing) : ______

(c) Find where the function is concave up and concave down.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (concave up): ______
ANSWER (concave down): ______

(d) Find the x coordinate all points of inflection. If there are no such real x, type DNE in the answer blank. If there is more that one real x, give a comma separated list (e.g. 1,2). MAKE SURE TO ENTER THE NUMBERS IN INCREASING ORDER.

ANSWER (inflection points): ______

3.(1 pt)
This is problem 21 Section 4.3 page 215.

Given $f(x) = 1 + 2x + 18/x$

(a) Determine where the function is increasing and decreasing.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______
ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (concave up): ______
ANSWER (concave down): ______

4.(1 pt)
This is problem 22 Section 4.3 page 215.

Given $f(u) = 3u^4 - 2u^3 - 12u^2 + 18u - 5$

(a) Determine where the function is increasing and decreasing.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______
ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.
Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (concave up): ______
ANSWER (concave down): ______

5.(1 pt)
This is problem 24 Section 4.3 page 215.

Given $f(x) = \sqrt{x^2+1}$

(a) Determine where the function is increasing and decreasing.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______

ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* . If the function is either not concave up (or concave down) then simply enter *NONE*

ANSWER (concave up): ______

ANSWER (concave down): ______

6.(1 pt)
This is problem 28 Section 4.3 page 215.

Given $f(t) = t - \ln(t)$

(a) Determine where the function is increasing and decreasing.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______

ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* . If the function is either not concave up (or concave down) then simply enter *NONE*

ANSWER (concave up): ______

ANSWER (concave down): ______

7.(1 pt)
This is problem 30 Section 4.3 page 215.

Given $f(x) = e^x + e^{-x}$

(a) Determine where the function is increasing and decreasing.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______

ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* . If the function is either not concave up (or concave down) then simply enter *NONE*

ANSWER (concave up): ______

ANSWER (concave down): ______

8.(1 pt)
This is problem 32 Section 4.3 page 215.

Given $t(\theta) = \sin(\theta) - 2\cos(\theta)$ on the interval $[0, 2\pi]$.

(a) Determine where the function is increasing and decreasing.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* .

ANSWER (Increasing) : ______

ANSWER (Decreasing) : ______

(b) Determine where the function is concave up and concave down.

Note: Express the answer using interval notation. If the answer includes more than one interval write the intervals separated by the "union" symbol, U. If needed enter ∞ as *infinity* and $-\infty$ as *-infinity* . If the function is either not concave up (or concave down) then simply enter *NONE*

ANSWER (concave up): ______

ANSWER (concave down): ______

9.(1 pt)
This is problem 6 Section 4.4 page 227.

Evaluate the limit

$$\lim_{x\to+\infty} \frac{7000}{\sqrt{x}+1}$$

______

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

10.(1 pt)
This is problem 8 Section 4.4 page 227.

Evaluate the limit

$$\lim_{x\to+\infty} \frac{x+2}{3x-5}$$

______

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

11.(1 pt)
This is problem 10 Section 4.4 page 227.

Evaluate the limit

$$\lim_{x\to-\infty} \frac{(2x+5)(x-3)}{(7x-2)(4x+1)}$$

______

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

12.(1 pt)
This is problem 12 Section 4.4 page 227.
Evaluate the limit

$$\lim_{x\to-\infty} \frac{3x}{\sqrt{4x^2+10}}$$

____

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

13.(1 pt)
This is problem 14 Section 4.4 page 227.
Evaluate the limit

$$\lim_{x\to+\infty} \frac{x^{6.083}+1}{x^{\sqrt{37}}}$$

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If needed enter ∞ as *INF* and $-\infty$ as *-INF* .

14.(1 pt)
This is problem 16 Section 4.4 page 227.
Evaluate the limit

$$\lim_{x\to3+} \frac{x^2-4x+3}{x^2-6x+9}$$

____

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

15.(1 pt)
This is problem 20 Section 4.4 page 227.
Evaluate the limit

$$\lim_{x\to0+} \frac{x^2}{1-\cos(x)}$$

____

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

16.(1 pt)
This is problem 22 Section 4.4 page 227.
Evaluate the limit

$$\lim_{x\to+\infty} \frac{\ln(\sqrt[3]{x})}{\sin(x)}$$

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In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE* .